ActInf Livestream #022.2 ~ Psychophysical identity and free energy

Livestream | Psychophysical identity, free energy | 1:51:50

okay
hello everyone we are live welcome to
actin flab live stream number 22.2
nice repeating digits there we're here
with alex kiefer and others we're going
to be having our second
follow-up discussion on the paper we've
been discussing for the little
last bit psychophysical identity and
free energy
it's may 25th 2021 and this will be
a great discussion so thanks again for
everybody who's
joining live as well as watching in
replay if you're watching live
especially feel welcome to add a comment
in the live chat that we can address
welcome to actinflab everyone we're a
participatory
online lab that is communicating
learning and practicing
applied active inference you can find us
at the links that are listed here
this is recorded in an archived live
stream so
please provide us feedback so that we
all can improve our work
all backgrounds and perspectives are
welcome here
and we'll be following good video
etiquette for live streams
we are here in the second of the two
discussions in late may 21
and happy to have alex joining us for
both of these discussions which has been

great today in 22.2
quick correction we will be just trying
to learn and discuss
enjoying these conversations seeing
where they go enjoying what questions
bring up and also
carrying on with a few of the topics
that we knew that we wanted to cover
from last week and with that
being said we can jump right in so we
can go to
the introductions and we'll just um
maybe go around introduce ourselves say
hello
and then end with alex and alex would be
happy to hear your thoughts or
reflections in this past week
so i'm daniel i'm a postdoctoral
researcher
in california and i'll pass to dean
hi folks i'm dean i'm from calgary and
uh yeah i'll pass it to adam
uh i'm adam i'm a postdoctoral
researcher
at the johns hopkins center
for uh psychedelic and consciousness
science
and stephen oh hello i mean stephen i'm
in toronto
i'm a community development practitioner
and applied theatre artist
and i'm doing a practice based phd at
the moment and i will pass
it over to alex
i think you're the only alex right now
okay okay i wasn't sure there's a fellow
jester
i watch it um oh so i'm alex kiefer

um thanks for having me back again to
talk about this paper
um i'm a philosopher i'm
currently affiliated with monash
university and i'm also doing more
applied work
for nested minds solutions uh it's a
kind of active inference ai startup
cool um there's a few ways we can
start but why don't you feel free to
give any opening yes and then we can
ask questions we have some other slots
right so you asked about my reflections
over the past week um
i haven't reflected a whole lot on deep
themes um i
did uh go through so
in the point zero discussion there was a
um
sort of a question about how we get from
two point one equation two two one to
two two
um with the flipping of the uh p's and
q's
so that strikes me as like not the most
central theme we could
spend time on but i did uh just spend
some time writing out the derivation
if that's something we can get to at
some point um
but in general like i guess my feeling
is i think
last week was really fun and we kind of
it was very far-ranging
um i i think it would be fun to like
to drill down a bit more on some of the
specifics this week but
you know it's kind of also whatever

people want to say so
that's that's what i'd be interested in
doing thank you alex welcome
blue if you'd like to just say hi we're
having intros and then we'll sort of
walk through some
jokes and questions good morning i'm
blue knight i'm an independent research
consultant based out of new mexico
cool so there's a bunch of places that
we wrote down
to jump in um maybe actually we could go
to dean's three functions walk into a
bar
so it's slide 11 and i'll have it up
large on the stream
so dean why don't you unpack what you
were thinking here and
let's get some initial reads to get the
contrastive ball rolling
well essentially what the the point two
aspect of this
process does is it takes um
[Music]
basically what we've introduced as as
as the topic uh tried to gain some
interpretation around it and then
so what does that mean what does that
turn into and so
um one of the products of that
for me was uh i kind of went back and
reviewed the um
the 22.1 and one of the things about
what you mentioned alex was the
explication of a simple idea
and i thought that was really important
and so
[Music]

what can that what can that turn into
that simple idea
that was the the three functions walk
into a bar piece
and essentially spoke to what scott was
talking about last time in terms of
paradoxes and what do those mean and
when do we model and when do we follow
but i thought the bottom part was
was an important piece in terms of you
mentioning
uh that you saw continuities
and i wondered if you were referring to
cynicism
and that idea of how
we take discontinuities and turn them
into continuities
man that's that's really interesting so
i had never heard of cynicism
i keep coming to pier paris through like
strange avenues that were not planned
um i never consciously thought of it
this way but this kind of nicely
summarizes like how i see things
and and the the way i've just been sort
of forced by experience to approach
philosophy
it like like like i i found it
difficult or impossible to draw to draw
boundaries that would create
discontinuities
between categories quite generally so
you know that gets into things like the
sororities paradox there's plenty of
examples of this but
i think it's it for me it's much harder
to find a discontinuity than to
uh yeah than the opposite okay

can you maybe just walk us through these
three functions
walking into a bar what what is the
narrative here
and where does it sit in relation to the
paper
you're asking me yeah yeah okay so so
essentially
if we're if we're gonna if we're gonna
talk about contrastive divergence which
i think is
a critical path to understanding how
we move down a gradient
and arrive at some sort of um
generative model i think that there are
three functions that
essentially exist in that it's an
inversion
process which i again i'm not going to
go into the
the the the reads on that there's an
identity piece which
is spoken to explicitly in in alex's
paper
and then there's the contrastive diverse
contrast of divergence piece
which enables a person to remain
balanced despite the fact that they
don't have a particular answer that
there are hidden states
and i think what a lot of what was
being spoken to last time uh scott
mentioned some things blue
uh stephen was things like scale
which don't necessarily we don't come at
it with
with the same perception as to what
we're looking at

some of us look at things a little bit
on a grander model and some of us look
at an on a granular
model and i think that the part
of being able to fit that into a a form
like the joke the three three guys walk
into a bar or whatever
kind of speaks to the fact that um
if you think that this is going to be
about stability
you're going to get interrupted by the
change aspect of it if you think it's
going to be all about the change piece
you're not going to be able to realize
any sort of conceptualizations because
conceptualizations require stability
so you have to kind of be able to play
in both
fields at the same time and that's hard
and i think that's why
alex's paper probably knocked me down
about five times when i first read it
because
you actually did alex try to somehow
pull together
the ideal idea of stabilizing something
even though things were
pretty choppy and and rather dynamic
so i don't know if this is an example of
where the part two
of us discussing your paper goes but i
think it's
it's interesting especially in the
context of cynicism and your
idea that we're constantly trying to
move from the unknown
to the known as opposed to the
traditional this is what i know

and i'm moving into these unknown spaces
that reminds me of then anyone alex or
anyone else
reminds me of sort of two complimentary
ways of thinking about education
with the tabula rasa the blank slate
that inscription is put into
it's like filling the cup with knowledge
or inscribing into something that's
templateless and then the alternative is
to burn away
ignorance and so it's almost like a moot
whether we're moving from the known
into the unknown or the unknown towards
the known
they're they're are left in our right
hands of education
right yeah yeah this is really cool i
mean it
it seems like um uh
dean like you have you have an act for
like reading
reading things in a sort of very uh
deeper almost allegorical way we're like
i'm talking about it
yeah yeah i read charles saunders pierce
two decades ago
before it became popular right so
yeah so so i think it's interesting and
it's it's all um
these connections are really there but
um i just just to to comment on this for
a second um so
so so i do think that there's something
about just the
how ubiquitous sort of dynamism is
and all this and how there's in a sense
there's no stable process that we can

latch onto here but
when i was talking about contrast of
divergence my main point was just
um well look that's i'm thinking of that
very narrowly as a machine learning
like an unsupervised learning algorithm
right which i i know you have that in
mind too but you're also taking a
broader
picture um so
um my point in contrasting contrastive
diversions to
uh simulated annealing was uh
we can suppose that the brain is running
something like contrastive divergence
where it's trying to it's trying to
reach a lower energy
state um but you know
the way that algorithm works of course
you don't need to
actually run right the basic idea
um is that you don't need to
so so okay sorry let me take a step back
for a second
um right so like one way of doing
this sort of unsupervised learning would
be to to take the distribution over
hidden states in a neural net that's
induced by an input
and uh try to
uh lower the energy of those states
while raising the energy of the
fantasies that the network would produce
top down right um and so
the ideal way to do that would be to run
run the network until it reaches its
equilibrium
distribution and sample from that when

you're doing the negative
the negative phase of this of um
increasing the energy of the or the
therefore lowering the probability of
the
fantasies um contrasted divergence just
takes a shortcut and just runs this for
like one step or two steps or whatever
um so i just want to make that clear it
i don't know if it's clear for everyone
in the audience who hasn't looked at
this
deeply um that that's
the the specific contrast that i wanted
to draw there was just
was just between that and a process like
simulated annealing where if you
if you run it to the to the end of the
process you end up reaching the lowest
possible energy state
um so the brain i guess my point is the
brain doesn't necessarily need to be
settling to
and this does i think connect directly
to your point it doesn't need to be
settling
to a completely you know stable state or
even
um uh any particular like metastable
state it just has
to be there has to be this gradient
right
that is available um but i do think that
there's a deeper point here
right so the nests or the
non-equilibrium steady state that
um organisms are are
approaching uh as they as they do this

variational inference
thing is not any kind of final
um you know it's it's not it's not a
static
state like some kind of uh
ultimate equilibrium right well going
down a set of stairs
the one the one steady thing is that i
remain balanced and not go tumbling to
my
eyes right that's the one steady thing
the one balance thing
and then there's the gradient itself
that's why the there's a next slide that
kind of speaks to the
that piece of it and i that's all that's
all i basically wanted to bring up
because i think
and i think the cynicism piece embraces
or
entails all of that there's a there's a
dynamic piece
and there's a balancing piece and you
and you want to be able to
sort of address both that's that's cool
yep thank you dean stephen and then adam
there's interesting questions here for
pragmatism
and social constructivism you know in
terms of how these ideas
roll out into how we might talk to other
people in the world
around projects and stuff
because of this implication so i i
wonder
how you you would bring in
this kind of embodiment of that so
um if the brain is an organ

that is doing expected more of an
expected free energy with the nervous
system
to help to manage the movement
of the organism and the organism overall
has got that variational free energy to
maintain
homeostasis you you you've got this
you've got this energy balance piece but
you've also got that need for entropy
so i wonder how with how this can inform
us
just like people when we do i don't know
if you know about the alexander
technique but it's when someone's
when you think about the body and the
skeleton we think about it as just the
bones but actually it's really
everything's held together in a muscular
sort of
watery yeah and
and and with the tendons and everything
so you there needs to be
some noise in there that that entropy
piece needs to be there when you're
going down the stairs and that
you can't be frozen you're full
yet you need to have some of that energy
equilibrium because again you're full
so there's i wonder how if we were to
talk to someone out there about their
life and their world
um walking down the stairs what what
would we translate this
in their language
alex if you want to give a thought on
that
i'm thinking about it um i'm not sure

beyond saying that i do think that
there's a
balance of two sort of uh
forces or quantities here always um
but i don't this reminds me actually i'm
not gonna try to put words in your mouth
but this reminds me of
uh adam of your your remarks about
free will in relation to degrees of
freedom and entropy and stuff
um so
that's a segue but it doesn't have to be
taken up
cool yeah it reminds me of stream of
consciousness or walking downstairs you
can't just
stop there statically can't stop
anywhere along the staircase
and you can't halt the stream of thought
so what is happening that's allowing
time t plus one to be generated from
time
t the state only has to move by one
moment at a time it doesn't have to do
the ultimate um steady state to the
lowest possible energy
so adam and then anyone else raising
their hand
um i reread your uh paper last night and
it's even better the second time um i'd
like to actually talk to you later
about like getting into the weeds about
different um
uh like mappings on the neural level of
like between
the thermodynamic and the informational
um but yeah
before we move on i was wondering how

you'd feel about
um like a multi-phase
uh or a multi-part description
um of neural architecture where like
there's some aspects which might be more
annealing like
in some aspects that might be more
contrastive like
so like you could like for instance tell
a story of like some sort of um
uh predictive workspace where like you
can think of it as like that the model
selection is like iterative annealing
potentially in terms of some of its
aspects
um but then like it's just like around
like you you could think of like like
like the hain style ignition event
as you're generating this this big high
temperature complex
that then settles down and selects the
model but this would just be like one of
the things that's happening like
something contrastive could be like
the thalamus like maybe like
orchestrating different
cycles and comparing them more or like
the hippocampal system and that could be
like doing your contrast
like would that be copacetic or with
that right
yeah no that sounds that sounds awesome
i mean in general it seems
consistent with like the way i've just
i've learned to see these things you
know just just paying attention to all
these different sources of evidence it
seems like

um yeah this kind of
um it's it sounds right to me i guess my
only
uncertainty is like the reason i focused
on um on this contrast was in part
because
i wanted to argue for as you know for an
identity between
the the energy free energy in the
thermodynamic sense and in the
variational
sense relevant to variational inference
and so
uh i needed to i needed to make explicit
the argument that whatever's going on
even if if there's sort of a
um multiple time scales in which this is
happening or or if there are different
parts of the system that are
um in effect being
annealed to different degrees uh that
the whole thing it's not possible to
think that the whole thing settles to
like
you know the lowest possible energy
state
while it's alive so anyway i don't mean
to like deflect
i think what you just brought up is
really interesting and you probably know
more about the relevant you know
neuronal mechanisms that i do so i can't
like
critique that idea but it sounds as he
said copacetic
yeah adam you want to add something
there oh i mean yeah and i wouldn't want
to like distract him like the point of

like
if we you actually you have to not be an
equilibrium you have to have this like
open endedness and dynamics to keep the
process
going um but i and i i don't know
like how productive like sometimes like
i'll squint it like
different machine learning algorithms
and like you know
push them a little bit further so like
you could think of this entered a
bayesian model selection it's like a
kind of annealing
but that could like not be productive
potentially even though aspects of it
could be described that way
yeah well that's a good i mean it's kind
of a good way of framing it like i think
this goes back to this synchrotis
cynicism
thing like you i think you can probably
always redescribe
something in terms of something else but
it's a question of how useful it is
and um and i never exactly know where
the boundary is right is
what becomes just kind of a fanciful
like philosopher's trick versus what's
useful
but the thing that you're suggesting
sounds like it would be a useful way of
thinking about things
and and when i contrast contrast of
divergence
and annealing when i think of annealing
i think of dna
in the test tube as you turn the

temperature down so you come to a final
frozen state that's like a fixed
crystal essentially whereas the idea of
contrast of divergence
it's more like cybernetics it's like a
guiding in a navigation
on a flow already and so even if you
didn't have any divergence at all
between your north star
and your header where you're going and
where you want to go
you still would be moving and so in that
sense even if you're trying to minimize
the divergence
there still is a sense of movement
whereas in simulated annealing
the idea that there's no difference is
more associated with a crystalline or a
static state
so they have very different sort of
telios
almost and for dynamic systems that
person going down the stairs
stream of consciousness action selection
in an uncertain world
that feels more like the cybernetic
navigation
rather than the dna in a test tube
though those are just
some aspects so stephen
and then adam
that that idea of um how things
keep going and the dynamics um
i'd be interested in your thoughts about
um
how that might play out in in behavior
or in the world so i could imagine in
the evening being a little bit like if

things are trying to hibernate there may
be a state
where they can go closer to like like
glass glass gets annealed basically
that's how you plunge into water and it
it holds that liquid state even though
it's colder than liquid
but the um the question
i have is is how do you
reconcile some of these um these these
these
questions around where we can get caught
in loops
for instance you know that um casper
hesp talks about
depression where you're seeing
a lack of affordance in the environment
and that could be confirming your
predictions
you can get kind of caught in a loop and
some of those
challenges in computational psychiatry
are they're sort of low energy i
actually haven't got an answer
it's made me think but they're sort of
low energy states but also like
looping conformationally states you know
where you get a reduction in free energy
relative to some
some sort of stick in the sand that has
been established for the priors
and the dynamics and then it kind of
gets stuck in that you know be it
you know hyperactivity something that's
outside of where the organism
want to be but it's still going there
and i know if that's
quite the same as um going into the sort

of a low energy equilibrium you know
going to it's not necessarily going down
in energy it's sort of
getting caught in a loop though of
variational free energy
yeah that's that's really interesting
actually um
sorry it was someone else wanted to oh
no alex you're going to hang up
and then after your thoughts will go to
adam yeah so i mean my
my my intuitive reaction is that
is that yeah something like that happens
in depression there's a sense in which
it's a lower energy state um
in my my paper which is kind of like a
sophisticated argument for a really
simple-minded idea i want to identify
that
those two things and say yeah it's
physically a low-energy state as well
um but it's a local one so uh
again this is my intuition at the moment
i don't know if it'll be borne out or if
it'll work
but it seems like sometimes in
depression maybe we reach like a yeah
like a local energy minimum in a certain
part of the network or something
but it's not part of an overall optimal
state for the organism when it comes to
behavior and things like that
uh it's really interesting to think
about how that happens
yep thanks adam
um uh actually to speak what uh steven
just mentioned um
there's a um within like the psychedelic

uh research communities there's uh
interesting models based on um
therapeutic change through annealing
where mike
johnson at the quality research
institute has something called like the
neural theory
or theory of annealing where it's the
psychedelic state increases the
temperature helps you to break out
and then you can crystallize in your
regime and uh robin hart harris is
described in very similar terms
um one uh thing about like
generalizations of annealing
uh that came to mind um when dan was
talking was
um uh doug hoff setter and melanie
mitchell
um with their copycat architecture um
they said it's different from annealing
but they had something that was kind of
like ant colony optimization and
kneeling
where there would be a temperature
parameter for when your analogies
weren't fitting
and so you can think of like some sort
of contrast of bits saying like how good
is your like matches how good is like
these analogical matching like discrete
comparisons
and then this parameterizes the
temperature of a continuously adjusted
annealing process that doesn't settle
and and more recently really
interestingly is um
um well from the beginning the language

was completely like work spaced up
and uh more recently uh melanie mitchell
has like explicitly drawn a connection
to the global neural workspace theory
and so there could be like a generalized
annealing that provides
that it has a contrast a bit and an
annealing
generalized annealing bit that does give
you like two kinds of like
psychophysical identity mappings maybe
and a total a total speculatory note
would be
sometimes we have you know accuracy
minus complexity or pragmatic versus
epistemic gain
maybe some decision making approaches
are able to hold both together
and maybe even rebalance between those
different modes
dean i'd like to actually go to 12
and hear a little bit more on what you
thought about contrast of diversions
before we head
to the next topic so what
did you see or find interesting on
slide 12 what brought you to that slide
well
i'd actually looked at stuff from oliver
woodford before
and so that was kind of why again i was
i was sort of i was connecting dots in
my own head
and it's interesting because the parts
that adam and
steven just brought up in part in part
of our conversations
off offline one of the questions i asked

daniel and blue was around the idea so
you have the physical math
and you have the statistical math and
somehow they come together and make
babies
and that's what we kind of have here and
so
what is this what is this project and
all i
liked about the not to
not to discount or put aside the
simulated annealing piece
but to give what the what i think
happens as a for example when the person
goes down goes down the stairs under
control
as opposed to falling to their death you
know lowest energy state
i thought that this essentially spoke to
it by just giving a nice metaphor in
terms of
shining the the torch beam in our chosen
direction of travel
which allows us to see the lowest point
in the field in that direction which
i think alex was again addressing i
don't think alex you
meant to make this such a a central
point of your
paper but i i thought in terms of um
trying to explicate the simple idea the
fact that you
didn't just focus on the fact that
there's a gradient but
how does a person move down that
gradient and not
lose control was kind of what i wanted
to bring forward here

and in the notes at the bottom which i
can't see on my screen
there's a really nice explanation for
what
contrast divergence does in terms of
being an assistant
that's why i want to bring that one
forward cool
alex if you want to give a thought on
that and then stephen
i don't have much of a thought i mean i
think it's it's a really cool um
angle that you have on this and it's i
think it's
again it's it's something that's there
even if it wasn't my intended focus
and it reinforces i think some of the
themes that i was focusing on
thanks stephen
so what i think is quite useful
and i'm kind of be sort of bringing this
idea if i was trying to explain this to
an arts practitioner or someone in it
does
you know psychotherapy or something
how to explain the implications and one
thing i think is important
here when you talk about that energy
minima if i jump to the bottom of a hill
i metaphysically i might know that i'm
at a lower energy
potential energy overall but actually i
probably get a higher energy when i hit
the floor for a while
right as i start to get stressed and
perspire and
assuming i'm not just completely
squished to oblivion

so you're in this interesting you know i
suppose we do that on roller coasters a
little bit
but we we there is there's what we
are inferring about our states isn't
what it is because we don't directly
experience being at the lower energy
state
we experience what our sensorial
states enable us to infer
about the results of our actions so
and then from that we compare that to
our expectations
based on the stick in the sand or the
big sticks in the sand which is our
phenotype
which is what we can then use as
something to anchor against so i'm
wondering
that that has an implication about how
people think about the world because of
these types of ideas
you know there's there's a sense that
they are
connected and interconnected at the same
time they're always
separate in terms of being able to touch
something
because they're not touching something
they're getting sensory data
which makes them get that allows their
model to predict what they've just done
but they didn't actually touch in the
way that people might think they've
touched
so i'm just putting that out there as
some question
yep alex go for it yeah that

so there's there's this piece um of what
you keep
doing stephen you keep trying to make
this like put
like put all this in touch with like the
body in the world and stuff and i'm like
no let me just stay in my internal model
and not worry about that
but so i don't know actually i don't
know if i if i understood the thrust of
just like the last couple of
sentences but i think the point that
there's
the fact is we don't we're not yeah we
don't just want to like
you know leap to the lowest energy state
possible um
become what may right it's it's much
more about the process
and um i think what you said about the
this thing being anchored in the
phenotype and you know the priors
in general is is is really important um
one one another paper that i wanted to
write or i don't know if it's a paper
maybe it's just a blog post or a tweet
or something but
um it's just on the fact that uh
yeah so like you always have to bear in
mind when you're talking about this
energy minimization process that
this is all relative to a phenotype
which itself embodies some amount of
energy
right and so in fact you might even say
that pheno the stable phenotypes
are some kind of like optimal tradeoff
between accuracy and complexity

right just in themselves um so there's
this force that's sort of pushing up
from below which is just the
having a phenotype complex enough to
predict anything and you have to work
within that
like those boundaries so i don't know if
that really addresses your point because
again i totally missed the embodiment
part just like
flew by me but it's it's nice
just one point and then adam so i really
like that example of
a person or a body in motion going down
the stairs and the difference between
the controlled walking down the stairs
um or maybe even using an assisted
mobility device
and then falling down the stairs and
both of them
can't stop at one moment whether you're
tumbling or whether you're walking you
still can't stop there
so it's always dynamical and you are
going downhill
and you are converting your potential
energy into
you know going to a lower energy state
so those two
are good ways to think about where does
bodily control and motor coordination
which we'll talk about in 23 soon with
embodied skillful
performance but let's think about a
mountain climber
that's somebody who's actually
contrastively diverging
uphill and so that shows us that it's

not just like
um the dye is cast and then whether you
fall or control on the way down
but there's a way in which going to the
top of the mountain
is like going downhill or annealing or
reducing the contrast and divergence
relative to your expectations of where
you want to be
and now you can imagine if the water
level is rising going uphill is the
survival strategy
so we want to have a way of describing
where does control
and bodily control come into the picture
going downhill
and then even generalizing beyond just
bodies with mass
falling in a controlled or uncontrolled
fashion towards
the kind of cybernetic planning that
could actually result in a mountain
climber for example going a little
downhill and little to the side and then
making their way up the mountain
so alex then adam then stephen
yeah yeah now that's really it's a
really good point to bring up especially
in this context of
you know of the thrust of trying to
simplify things by identifying physical
and
you know psychological things like no
sometimes you want to climb the mountain
so i mean my my my immediate way of
trying to summarize how that would work
is just well you've got
a strong prediction um that yeah that

you want to be at the top of the
mountain right that's your sort of
that's your sort of desired distribution
um it defines your desired distribution
over observations
and so that's gonna it's gonna drive you
to climb the mountain but i think
that the way it does that is proximally
is that there's going to be an actual
i would say an actual physical uh
disequilibrium
in the brain right that's caused by the
discrepancy between your predicted state
and the state that you're inferring from
your observations and
lowering that uh you know that potential
uh is what drives the action of actually
climbing the mountain so
um yeah
cool awesome adam and then stephen
head's spinning a little bit but
hopefully this works
because like up down um but i'm
wondering in terms of uh something like
uh depression um uh
could we think of it as like a very
high local minima
and like if we thought of like what we'd
expect so i guess okay
what is it that's being predicted at
which level of
the degenerative model um like what is
it they're being
is what's what's being predicted like
and actively
like between the original niche and
what's being predicted
over the you know whole and like skin

encapsulated organism the
brain and then subsystems there could be
some ultimate they have to come to some
kind of harmony
but they they could diverge but um the
idea would be like thinking
of um uh
a landscape that like when the energy's
high it might be
more jagged and you might be more likely
to get stuck at like
uh local you're stuck at this local
minima
and you want to get to the the a really
low one
and so you could think of something like
uh niche construction
as like looking for a catalyst to like
try to like
smooth out like make a more navigable
landscape or you could think of like
the psychedelic would be like in turning
the heat so i'm thinking of the
difference between like a
a thermodynamic versus a kinetically
favorable chemical reaction so like
you might like yeah it's theory if you
get from there there you come up ahead
with the anthem
with the enthalpy but you can't get
there it's not kinetically favorable
and so like the different things we're
doing is trying to like
uh find ways of getting kinetic
favorability for thing for this
thermodynamic favorability and bring
those into alignment
very nice atom you can build that bridge

so you don't have to forward the river
or swim across or you can just pray for
the quantum tunnel
to appear so i do
stephen and then dean
wow okay there's something very
quite important here i think because
okay we're minimizing this energy
gradient okay that's fair enough
particularly for robots which are using
reinforcement
approaches and they're basically working
on energy gradients and then you've got
the idea okay you can work with
variational
approaches i more like this
gibbs free energy where the reaction can
slightly have entropy
but the big key thing as well if you're
going to take your agent is
that where at the level of the chemistry
is the chemistry is all non-equilibrium
chemistry so the whole of the gibbs free
energy is built on equilibrium chemistry
here we've got a non-equilibrium
chemistry when you get to the level of
the organisms and cells and
everything so i think what's an
interesting
point here is so you're going down the
stairs you're trying to fall over you're
being a rock climber
so overall there's this energy piece
around how the agent moves
but for the agent to move unlike a robot
where you might have little pistons
the actual cells and that are operating
at this non-equilibrium state

and quite how different actually this
gap between
these attractor basins versus
energy basins which we normally think of
i actually don't know
i don't know i'm interested to ask carl
fristan that like what is
how how much does it shift things to
have and
basins are to do with the tractors as
opposed to just basins to do with energy
like equilibrium energy gradients um
and and that's the thing that's
different in terms of what's really
going on when i'm climbing a mountain or
that
is all my muscles and that at the actual
level of making things happen a swarming
and trying to enable this kind of
biological process to happen and okay
from a system perspective i can see the
energy being less for me falling on the
ground
but for my muscles to move and
everything
um they're they're operating at this
non-equilibrium
process and which will have energy
impacts but also
as some other types of dynamics which
you don't have you have this
non-equilibrium steady state dynamic
which you don't get in traditional
chemistry
so there's something there i just
thought i'm not saying i expect an
answer from that but i think it's a
it's an important point that seems to be

a gap it just it really
shun out for me there nice thanks steven
it's kind of like if it's a ball
at the bottom of a bowl then it's it's
that's the annealing
it's the crystal state with the ball and
the bowl but then if you have a neuron
that's holding a resting voltage that is
being actively maintained
with an enormous expenditure of energy
but it's its own kind of
bowl that it's converging to whether
it's above or below
its set point it's still doing a type of
convergence but it's a dynamically held
state
and you're absolutely right that
sometimes the lower level
entity or the entity which the larger
one is composed of
they have their own active dynamics even
when the top
level system is at rest or appears to be
unmoving
so dean and then we have a few other fun
topics that we'll get to
well i'm gonna see if it's okay to
ask ask alex to jump to his third rail
at this point because i'm really really
curious about the modified ramsey
sentence
uh aspect of his paper and what i was
hoping he might be able to speak to a
little bit about is
not just what that what the ramsay
sentence is talking about but
the implications in terms of different
time intervals and what that

role that plays in us being able to
understand how
the modified ramsay sentence works in
your paper
perfect i switched to that slide um 44.
so yeah i was definitely curious about
that as well alex what does the
ramsay sentence or this entire area of
formal logic how did that come into play
yeah so right i kind of trumped up the
ramsay sentence thing in the last
episode but um
it's kind of silly of me because there's
not a whole lot to it
in a way but then maybe that's just how
it seems to me from
the perspective of where i'm coming from
but
the reason this is in here is that the
whole identity the philosophical
mind brain identity piece of this was
based on i was mostly inspired by david
lewis's approach to that topic
and so there there are a couple of
influential approaches approaches
there's lewis's and there's um
uh smart i guess the two big papers um
but basically uh what but this piece of
formal logic
is doing is just trying to capture what
um what a theory
says about the entities that it sort of
implicitly defines
or how you could get from um
this isn't all explicitly spelled out
graphically but how you could get from
just a collection of statements that you
might have sent to about

whatever the topic happens to be in this
case we're talking about mental states
from that to a uh
a theory that's um expressed in general
terms and that
again implicitly defines the uh
the entities that figure as the values
of these variables here
um so the right so i mean we could
briefly talk about this i guess is it
useful to go over this
yeah you guys yeah so um read out the
sentence
um or translate it or provide an example
sure um so right so i guess i'll start
with just the ramsay sentence so we can
talk about modified
versions in a second but um so the first
symbol here as you guys discussed in
point zero is just an existential
quantifier it says
there's some x such that whatever
follows uh the second symbol is a
universal quantifier over the variable y
so it says there's some x such that for
all y
blah blah blah and here t the predicate
is just literally like so say that you
had a collection of platitudes or
they don't i don't want to complicate
things just a collection of statements
like uh um when i see
uh when i see a cat i'm happy uh when i
think about happiness
uh i tend to think about sunshine things
like that right
just some examples um and so you just
can join all those just put an and

between
all of the sentences of that type that
you could think of in this domain in
this case it's folk psychology
um so then you'd have you'd have a
a uh a long sentence right it's a giant
conjunction now
uh that has some terms that refer to
mental states like
uh believe arguably see although maybe
some people interpret that
as a physical state there's definitely
blurry blurriness here at the edges
but in any case things like believe
desire think these are clearly mental
state terms right
so you'd have this long sentence that
has those terms and also other terms
and so essentially you just you replace
each of the the
terms in question the mental state terms
with a variable and what's left over is
the predicate
that defines what the sentence is saying
about those
the values of those variables right and
then you just append these quantifiers
to the beginning um just to sort of
spell out
um what your what kind of claim you're
making um about
the states in question namely there is
such a
there is something such that it relates
to all other things in this way
um and so that's that's the basic ramsey
sends it's just a um
it's an abstraction that allows you to

um to spell out in formal logic and
first order
practical logic uh like what a
collection of sentences says about
something
um and the and the last piece here is
the so there's the if and only if the
three bars
um that's the resellian uniqueness
condition so this goes back to
russell's theory of definite
descriptions so like i got i mean my
start by the way was in sort of
philosophy
of language and stuff so that's why this
is in here i came to the
all this sort of statistics and physics
space stuff much later
um so so russell you know had this this
theory of definite descriptions
uh that he was trying to explain how you
could
have a meaningful expression uh like you
know the present king of france
um even though there is no such thing so
how does how do failures of reference
how do we philosophically handle
failures of reference and things like
that
um and so his his suggestion was that
um you can think of definite
descriptions like the present king of
france as
uh just saying well there is something
such that
it is a present king of france and
there's only one of those things
right so it's like a uniquely satisfied

description and so that's what this
end part does here it says this
predicate t which
encompasses all of the implicit folk
knowledge about mental states in this
case
uh holds uh holds of something only if
that thing is equal to x
and so one last piece that's important
here is that
i took x the the bold letters here x and
y to be
like uh vectors
essentially right so like these are
we're talking about many different
mental states
in parallel we do this uh but if you
want to you can just think of it simply
as being about one mental state but then
you generalize to
all of them so i don't know where
how clear is that what what what
it's it's let me try a uh
a little bit of a physical example it's
like there's the stream of consciousness
the stream of statements what you said
with an ant
i saw a cat or photons hit my eye
and i perceived a cat so one of those is
more of a physicalist claim
and one of those is more on a mental
side and it's almost like
we're going to draw a line around
the mental claims just so that they can
be
definitely interfaced into a formal
logic
even if the contents of that mental

experience
might not even have a reference in the
real world
or it might not be something that exists
in the same way that those photons
hitting the eyes
exist but it allows us to interface with
those kinds of statements
by agents in a formal logical framework
well i mean i guess one part i'd push
back on a bit is
i mean this sentence here clearly states
that
that this x exists whatever you take
that to mean right so it could be that
there's different criteria for existence
right in different domains or something
but it definitely said that that there
exists something such that it's related
to all the other things of this kind and
in the ways that are specified in the
predicate
so like yeah i think i think the
the the main i guess the main point here
i so i focused on the ramsey sentence
because people were interested in it but
like the main point is just
um that you can take all the things that
people
say about mental states as implicitly
defining them
in a sort of theory right and that
that's
uh that's um
and so lewis used this explicitly in the
context of trying to identify mental
states with brain states
right so just to i'll just say a little

bit more about this piece and then i'll
shut up for a bit but
like um if you have this network of
relations that's described by a sentence
like this
uh you can take that as implicitly
defining the mental states while being
neutral about their sort of ontological
character right so
um so it could be then that if you do
some
you know neuroscientific investigation
and you find hey there are these
you know states in the brain that are
they seem to bear the same relations to
one another that
these mental states that we were talking
about bear to one another
uh you know lewis says well then
because this the sentence says that the
mental states really are just the things
that uniquely realize this structure of
relations
well you've just empirically discovered
the mental states
um so it's a really cool way of
theoretically bridging
um something like psychology intuitive
folk psychology and and neuroscience and
lewis's point was
once you have this structure if you
could extract this structure from folk
psychology that's implicit in it
and if you made the empirical discovery
there'd be no further bells or whistles
whistles needed to say that there's an
identity between them because
that's just what the sentence says is

that the mental states are whatever
things satisfy this um
description very interesting so
dean then stephen then adam
yeah so so alex i mean as it's as it's
written out here it's
it's really stable and i'm i'm really
curious because i really don't know this
do different time intervals like really
really short time intervals
or longer time intervals if i've got a
day to go down the stairs versus
i've got one second to decide whether
the 96 inch drop
is less harmful to me than the lion
that's chasing me
do time intervals affect the stability
of this right yeah so sorry i neglected
that part of your comment earlier um
i think no is the short answer i think
that this is meant to be a static
a static sort of description of a
implicit theory
right which which will include
statements about short time scales and
long time skills
so i would say that that anything about
time scales
in theory should be captured by the folk
psychology that you
embody or that you have in mind at the
moment and that might change from one
time
moment to the next also like this
doesn't need to be a temporary static
structure
so maybe that speaks more your point but
um

but i think the possibility of of
spelling this structure out even for one
instant is the is the is the lever that
lewis uses to to draw his philosophical
conclusions
okay thanks thanks steven and then
adam
this question about
prediction and time i think like you
just saying
there may be a sort of time scale at
which things are just flowing
like what sort of speed is it that
things are just doing their thing i.e
cells are doing their processes but what
dean was talking about was
is longer time scales than that and i
think like you say there's something
different there that's a more
everything beyond a certain point has to
be imagined i suppose
anything that's not possible to do
variational free energy on
you've got to do expected free energy
and to do expected free energy you
basically have to imagine
some a big us imagine in a different way
to just our limited sense of imagination
like our body has to imagine at some
level
the expected free energy basically what
expected free energy is
so i think that's um that then does
bring in some interesting
ideas around folk psychology would you
say
that those mental states can ultimately
be action

and sensory states that are being
predicted so the mental states are
basically
the recapitulation or expectation on
sensory states and resulting beliefs on
action states
so they they could be seen as mental
states but actually there is a more
distributed
realization of that i'll be interested
if that that sort of
is something that could hold yeah i mean
i think
that the spirit of lewis's proposal is
also functionalist right so
so the the basic idea functionalism of
course is that the
mental states in the philosophy of mind
is that mental states can be defined in
terms of their
relations to inputs and outputs and also
to each other and it's i guess it's the
to each other
bit that distinguishes it from
behaviorism so um
but yeah i think the these implicit
definitions
that lewis talks about would rely
heavily it would depend heavily on
things like um when i believe that
there's a cat present i expect
certain visual appearances to occur
um and i'm disposed to act in certain
ways and
maybe you want to catch the the i i tend
to want to cache the actions and the
sensory stuff out in terms of
proximal stimuli and like motor effector

states rather than
distal things and extended things that's
just i
i don't know that's my my my um
i don't know if it's a bias or what
that's just the way i i tend to think of
these things but
um definitely
i i guess i would say that
the the relations between the mental
states though are just as important as
the relation to
uh the active states from the sensory
states um
when it comes to this kind of um
this kind of formalism um
i feel like i missed an opportunity to
it was a really interesting comment
actually i
i don't know if i did it justice but i'm
still thinking about it can i just
just add one little things like just
before yeah
okay very quickly but one thing that
comes to mind that i i
think might connect what you're saying
um you were talking about this and
and and so there seems to be something
there about
um the combination of ands
um and what's you know
so for instance and there was a dog and
there was a woman with the dog and were
in the park
and the dog was wagging its tail
and then there was blood on its teeth
when it smiled
and i'm like suddenly that changed it

right so the triangulation of the ands
flips once you suddenly the dog bears
its teeth and you see blood on them
right
so that links into what they do in
qualitative research
where they often triangulate to get
something which has got
trustworthiness rather than accuracy um
and i think there's something
interesting there about
um what you're saying in terms of that
my knowledge of
that field of linguistics is but i if i
can bring it back into embodied work i
find it easier which is what i'm trying
to do but
this and and and and that's that has got
some practical
applications i think in um in a number
of areas
definitely yep thanks adam um if you can
return yep
okay go for it adam hey making coffee
so um eventually i'd like to get to um
free will and actually uh loop around a
little bit to actually depression
as a case with respect to that but
um before we move on from ramsey's
sentences um
uh not to belabor it but so it is the
idea that
so you strip away the specific entities
being signified
you look at the syntax and then you're
saying it can the same syntax apply
in another system and then you have the
mapping that

this would be the map and so that would
is that the basic idea essentially yeah
if you take syntax in a you know in a
slightly broad sense
then totally yeah okay so i guess i'd be
wondering like um
like in terms of like active inferential
modeling would it be something like um
like with like the graphical brain like
you have like this forney factor graph
where you have like the continuous
regime down below but then you have this
like discrete regime up top
and that maybe something in like the
factor graph portion
you can look for like some sort of like
uh
homomorphism or isomorphism there or
like if it's the brain
you would do some sort of like you would
look at ensembles
and like describe them as like
attractors moving along some sort of
manifold
and then like when you core screen those
or throw a blanket around them or reify
them in some ways
if the syntax of their interrelations
matches people's
uh uh subjective reports of their
experience
then you would be able to actually
potentially identify like
physical sub physical and computational
substrates of consciousness is that
i mean yeah i have a bridge too far or
like that well there we're going
that's i think that's where where we're

going except uh
i never made the bridge all the way to
consciousness because
that's one thing that i i reserve the
right to not decide about i think your
work on this is really cool actually i
kind of suspect that like most of the
main theories of consciousness are kind
of
overlap a bit and there's a core that's
you know
that's sort of like uh getting at the
way things
are which i think you've written about
but um so so i want to just bracket the
consciousness thing for a second
although maybe that totally ruins it for
you i don't know
but like i think yeah but that's that's
exactly
the idea like i think you just expressed
it very well
that um if you look with
yeah you have to squint you have to do
some coarse graining but if you look at
the dynamics of the physical system
uh you'll be able to discover in the
ideal case some of the same
sort of syntactic relations that you
would get from the psychological
perspective and i think that's also very
constant with the way
you know connectionist modelers describe
their work sometimes so like
you know you think a connection is a
simple you know neural network made up
these
simple processing units uh you can

derive that architecture just by looking
at the brain and sort of
very much simplifying things or you
could
derive the same graphs in some case just
by looking at cognitive processes
and so you know it's the idea that you
have two perspectives on one thing
exactly
if you have a graph isomorphism then
that is what was being hinted at with it
being sufficient
to find that there's relationships
amongst mental states
like i always feel like i feel this way
before this other feeling
and then if you found that there was
some brain state that always preceded
is that the end point or is there still
more to explain in that context
let's actually get to another topic that
we highlighted last week
which was the sleep wake example so i'm
here on
slide 40 and we're considering a system
whose analysis is tractable at least
according to alex
and we have an agent whose behavior
as far as sleep wake is concerned is
defined by equation
3.1 so could we walk through what is
happening
in this simulation like what is the
agent doing what does the agent
consist of and then what does the
equation show
and then how does that relate to what
we're showing in this paper

sure so i think i think this is another
aspect of the paper that
um that was uh
maybe seemed to be a making a deeper
point than it was but
i'll just talk about this yeah briefly
so this was actually just meant to be
uh sort of a slight modification
of just like a helmholtz machine running
the wake sleep algorithm
so i don't know if i need to set that up
blue talked about that a little bit in
the point zero um
but the idea was if we just have a
generic let's we have a neural network
that has a generative model that's sort
of
encoded in its top-down connections and
a approximate recognition
model in the bottom up connections and
um
so so one way of doing this is to run
the wake sleep algorithm where you
basically generate
uh states of the network top down like
fantasies right
functionally speaking and then you
adjust the bottom up weights so that
they're likely to produce those
fantasies
and then you you alternate this with a
so that's the sleep cycle
you alternate this with a wake cycle
where you use it input
of the kind of sensory data you want the
model to learn to
generate and you do a bottom-up pass
and then adjust the the top down

generative connection so that they're
more likely to produce those
states that are induced by input okay so
um so all i was doing here was saying
uh look if we if we suppose just make
the assumption that this is a real
system we're talking about it's
ridiculously unrealistic right so this
that's why i said it's
uh it's it's something that's tractable
to analyze but it's also
also quite a far cry from you know
anything biologically possible
but it's not also it's also not
completely off the mark right so i think
there's a reason that the helmholtz
machine was kind of like
a seminal moment in the evolution
towards what we have now
um because it got a lot of the coarse
grained
um sort of conceptual structure really
down that i don't know if it had been
pulled together before
in this form anyway so let's suppose
that the wake and sleep cycles of that
machine
uh occur with equal probability then we
can talk about
probabilities of certain states
occurring in the network just
period right whether it's during a sleep
or a week cycle um and so if you just
grant that if you if you ignore the fact
that this is a
really only it ends up being an
approximation to an approximation right
so it's

it's um it's running it's uh wake sleep
um
ends up being sort of an approximation
to the em algorithm which is itself
not quite full variational base so
anyway lots of approximations but if you
ignore that
for the moment i just wanted to make the
point that you could talk you could talk
about the
you can thoroughly completely
characterize the possib the
probabilities of various states of the
system
and so you could use that to define this
this r here is just the probability
of a certain state of of the system
being in a certain state now here the si
is referred to like the states of the
individual neurons
but that's related in a simple way to
the overall state of the network
um so you can talk about the probability
of the system being in one of these
states
um regardless of whether it's running
awake or a sleep cycle and then
that gives you a if you have a
probabilistic description of the states
of the whole system
uh then you could you could use that
as as um
to define the the relevant terms in the
thermodynamic uh
free energy so the entropy and the
energy and such
so that's the basic idea but the the
part that you

that you were wondering about like what
is this um
system simulating what does it perceive
and such
that's all that's this example is
completely agnostic as to any of those
details
so once again this is highly highly
abstract and i apologize
no it's cool and what do you think this
model would
allow us to continue developing towards
or what does it show
for the argument that you put forth as
far as the psychophysical identity
thesis
right well i i don't know that it shows
anything in itself i think what was what
it was
trying to do was was was write down
a system in which the target would be
clear so like
the you want to be the case that this r
distribution um
[Music]
so so let me put this way so the r
distribution because it's
it's essentially made up of or it's it's
factored into
these wake and sleep cycles right those
have direct
um interpretations in terms of
variational free energy and inference
um and yet we we have a well-defined
distribution over the states of the
system
that we could use to describe its
physics so i just wanted to show that

there's one
simple case where where we can we can
write down a distribution that bridges
these two descriptions
and then uh then in the subsequent
paragraphs i try to
i try to argue that we can generalize
this beyond equilibrium and that's the
that's that's the very tricky bit and i
don't claim to have shown it by any
means i'm just
again i was trying to remove obvious
obstacles to this thesis being true
awesome so adam with a raised hand and
then blue or if anybody else wants to go
for it
um in terms of um wake sleep
uh there could be some uh potentially
biologically
realistic implementations of something
like that um o'reilly and
um has a paper uh deep predictive
learning where he describes
a model of the thermocourtical system
doing predictive processing that's
a very weak slate a wake sleep-esque i'd
be um
i'll send that to be curious enough if
you like if you thought that was like a
good like psychophysical identity
mapping there the other uh
thing would be um so
if
if we think of experience as being
the ramsification of experience with the
mapping would happen at the level of
predictions
um then and if the predictions

are the uh sleep part um
is there a sense that experience is
always like kind of paradoxically it's
always the dream
like the awake part is the stuff that's
just updating it but the actual thing
generating the experience
is the dream part like correct or yeah
no i yeah so i i have an unpublished
paper on this too on the
representational division of labor like
so people like in predictive processing
literature say things like
yeah we experience our expectations of
the world or whatever but i'm not
i'm not so clear that it's that way i
don't know um
i guess if you think that the
predictions are the carriers of the
experience or the
neural correlates of the experience or
something then then that would be
the case and the wake cycle would do
nothing but i think it's
i don't i don't see the reason to
embrace that that
that interpretation of of what's going
on here
um because it's awesome
no i mean i'm all about no no i'm all
i'm all about the idea that that
we experience sort of virtual models in
a way
right that's that's cool i just i i
don't interpret that in terms of like
it's just the
it's just the top-down driving signal
that leads to experience although

maybe i'm not really against that either
and i can see why you'd want to say that
um anyway i just don't think that
there's any inference from like the
helmholtz machine
to that like because and i would love to
look at this paper you mentioned by the
way on like
biologically plausible versions of this
because i think there's a lot that's
right about the only the reason i say
it's so
impossible is just that it doesn't have
any role for
top-down modulation of bottom-up signals
during perception
which was by design so that it would
work quickly
um but that's the main limitation that
concerns me
um cool thank you alex
steven with a raised hand and then
anyone else and also anybody watching
live you're more than welcome to
write a question in a live chat so
stephen go for it
just uh just to help me clarify so
this being an attractable
pro analysis so the key the key point
is that by going to stochastic
it shows in principle how
recognition and generative densities can
be used together
is that the kind of the point that this
shows
i just because i'm struggling i think
i'm just i'm not so good with the
formulas and to exactly get it but i

think the general point you're making is
that
you're going to stochastic algorithms
and therefore
you've got this potential for the
recognition and generative densities
which is basically sensory and
accurate potentially to be reconciled
and shown
even a very simple case would that
be a correct reason a bit i
i think so i i think definitely that's
um it's it's in large part right so that
the idea that we're dealing with a
stochastic
system is kind of important because yes
i want to claim that
or i want to advance the suggestion the
possibility that
probabilities are encoded by
probabilities so then you could have an
actual identity
whereas if you're just encoding like if
you're using like
i don't know the um
a scheme that's more like what carl
firsten talks about often where you
have sort of neuronal states that are
encoding the sufficient statistics of
distribution
then i don't think you could get an
actual identity because you just you'd
have a clear
representation that has a certain
um almost convention built in of how
things are represented
whereas i'm arguing for the possibility
of a much more direct encoding

and that does depend on this being
stochastic but the other part of this
that i forgot to mention
that sort of i think maybe crystallizes
the
point this is trying to make is that um
if if we're at equilibrium with respect
to the network if we're at the
um helmholtz machine's equilibrium
distribution then
um in that case if you do take rs
describing the physical probabilities of
the system
that you would use to specify the
thermodynamic properties uh
then you could show that in that case at
least the
variational free energy and the
thermodynamic free energy would be the
same
and then generalizing to non-equilibrium
of course is the hard part
but um that's that's that's more
concretely what i was trying to show
there
um yeah thanks
so i'm gonna go to a question from the
chat and then we'll go to adam
so cambridge breaths asks can this model
describe or map the transition between
focused attention
and mind wandering grateful for any
elaboration
thanks everybody
um anyone want to take that i
i don't know i don't know yeah i don't
know just
maybe the dream versus the you know

sleeping versus the waking
but then even within the waking state we
have a wandering phase versus a very
tightly focused
phase so there's sort of what is
happening
if that reminds you of anything i mean
i'd say if we're talking about this
model
talking about the helmholtz machine or
this like slight elaboration on the
helmholtz machine then i don't
think it would do anything that
sophisticated just because
like so as adam says there are
biologically possible ways of
implementing this kind of thing but like
anything is fine-grained as a shift from
attention to
mind wandering i think is probably going
to be left out of a model that's at this
uh level of sort of abstraction and
approximation that's my thought thanks
so adam then blue and then anyone else
with a raised hand or a question in the
chat
adam i'm not sure i can address the
mind wandering exactly with
psychophysical identity but in terms of
like
the mountain climbing analogy i'm
wondering if it's like
mind wandering is kind of like uh
exploratory hill climbing with
a little bit of quantum tunneling
something like that but
uh that's a that's a i mean what would
you what you think actually it might

loop around to the brain in an
interesting sense um
and that um there's
like if we think of like
lower levels of like a cortical
hierarchy as doing something
more like um a variational inference
uh via like continuous message passing
maybe with like some discrete updating
there too
um so
that seems to be like one kind of
implicit
representational regime there might be
a sense in which you're getting like
probabilities in terms of like the
attractor dynamics along this hierarchy
corresponding to like
probabilities of like a hierarchy of
nested events in the world and like a
deep temporal hierarchy
and the deep temporal hierarchy of like
the latent causes of the world
uh but there seems to be like potential
like as we like move inwards like
another level where these get um
re-represented into more like constant
like when you're into like deep
association cortex
it might be a different game and then
when you're getting like coupling with
like
the the campbell system uh now
it seems like you're getting something
almost like like explicit syntax
with like discrete semantics because
like
you can map like there's these pointers

going from like
the relationship among these like you
call like bump attractors
the hippocampus system or like these
place cells but they can be used to like
contextualize the whole overall
uh system within some sort of uh
structured syntax
um i'm i don't know like would
do you need something like that to have
the probabilities where for
um is that what you're looking for or
could there be more like this like
inactive hierarchy of uh like a
hierarchical boltzmann machine
that has a nested and typical cause
there and that those are probably like
yeah both do it or would only like
the re-representation yeah i don't know
i think a lot about this this issue of
re-representation
and how much you know if it happens and
what it's doing and stuff but um
i think uh i guess i need to understand
more
and this is totally just my ignorance
about the kind of
models you're describing with
hippocampus and such like um
i tend to think of um
i tend to think of of semantics in the
brain just in terms of like vector space
models um and so
uh i'm just curious i don't know if we
could talk about it now
for a bit maybe i don't know it's up to
daniel and others but like
um what is the explicit syntax in this

case it sounds something like something
that i
that would be friendly to to what i'm
trying to do um so i guess like uh
so this explicit syntax it's like still
heavily researched there seems to be
kind of like a gold rush
among different like ai cop companies
and like cognor labs of trying to figure
out
hippocampal system like deepmind this is
kind of like what started them
it was like kasabi's like making
discoveries about this
but the idea would be um that you have
this
graph structure that would be that
originally would have evolved for just
locations in like
uh meat space or physical space yeah but
then got repurposed for any kind of
space and so this gives you like
structured state spaces
whether you're navigating through the
world or you're navigating through like
a spatial topology right this would
basically the content
would be these um and so it would be at
the top
of the cortical heterokey um and it
would have pointers that could be
unpacked
in terms of like these more like modes
of an active engagement
there might be like another like
intermediate level could be associated
with consciousness
where you're getting like um more

concentrated
attractor dynamics with like potentially
a quasi-topographic relationship
to events in the world and so the
principles there
um uh graph neural networks might be
relevant there where basically you get
this
orders of magnitude greater
representational efficiency
by uh actually making the geometry
of the system over which you're doing uh
the deep learning
resemble the thing that's being uh
modeled so like
um there's like graph grid neural
networks that can be used for like
spatial modeling
or there's like graph mesh neural
networks that can be used for like
uh modeling of like someone's po like a
a pose of a thing or an object
and so in theory some things like this
might form at the intermediate level
and these might be the ones they're most
heavily coupling with this hippocampus
system which is giving you this
syntax of state transitions
right with like structured composition
that's
that's really cool can i comment on that
for a minute yeah um alex go for it and
then blue
okay um yeah so there's a lot there
that's that's exactly along the lines of
what i've been thinking about this i've
written a bit about
these like conceptual sort of grid

cell type things uh that it seems like
there's a lot of evidence for that it's
really awesome
i know like memento is doing some
research that's related to this stuff
um so that's okay if that's what you
mean then i'm totally on board like i
wasn't sure if you meant that or
something more language like
with like discrete tokens with
other parts but i mean that might be
encompassed by this as well
but it seems like uh people are finding
also like encodings of like the
hierarchical
uh structure of like the grammar of
language is also encoded
uh it's found to have a correspondence
uh right with like
the structure of the like they said the
relationships of these bump attractors
that's amazing yeah i was going to
mention also so you meant you gave all
these examples of graph
of neural networks that are sort of
isomorphic with the domain that they're
they represent um i was going to mention
recursive neural networks used for
language processing as well
the the the network topology matches the
syntax tree
or something like that so i that's very
that's very much along the lines of
where this is going thanks alex so blue
and then dean afterwards
so i want to kind of loop back around to
what adam said about dreaming because he
totally like took the words out of my

mouth but maybe um
also touch on the question that was
asked by the person in chat
about mental wandering so
is it possible so so alex your um
you know objection to the dreaming and
the physicality was
that there's no evidence for top down
control during the wake
cycle is that correct is that what i
heard you say um
sorry no i actually meant to say kind of
the opposite
i meant to say that um that the problem
with the weight with the helmholtz
machine as a model is that it doesn't
it doesn't model top-down effects
during waking perception right and
that's what i interpreted so
so okay so if it doesn't if it doesn't
model weak uh top-down effects during
the wake cycle um is it possible that
maybe
there are like overlapping helmholtz
functions happening right like so you
know your mind wanders off or
you're not paying attention and like
maybe you're consolidating
like you're you've got like one set
sleeping one set waking and there's
maybe like an overlapping construct of
this like wake
sleep learning cycle happening like
while you're out
mental wandering i'm consolidating you
know the information from
you know what i learned at breakfast or
or something like that

yeah yeah i mean i think there's i i
think there are there
are like like this model is adjacent to
like many
really interesting possible variations
right and slight elaborations so like
what you're talking about sounds to me
like in that vein
like i saw an amazing git repository of
just like
helmholtz machine variations right where
they just they just tweak certain
parameters like let's see what happens
if we
propagate a few steps down during the
wake cycle as well you know things like
that
um so i think i don't know so i don't
know if i fully grasp
like exactly what you had in mind but it
sounds pretty good
and thanks for i mean you said go ahead
oh alex go
oh no i was gonna say thanks for also
trying to address the comment and
because i felt like i didn't
wasn't able to answer it very well nice
and there's definitely
wandering in dreams and how is that
different than the wake
wandering and could that get at what the
difference is between sleep and wake and
could this model
maybe give us a bit of a wedge to enter
into that
discussion so dean and then anyone else
with a raised hand
so alex back

back to analogizing so i'm reading your
paper and
i can read something and i go oh my
goodness look there's the world's
biggest nickel
and then i continue reading and then oh
look there's the world's largest corn
cob
and then i have a chance to have a
conversation with you and you're like
saying to me well let's bring this back
into some sort of
proportionality so explain to me what is
the
2 carat deep diamond that you were
trying to
allow readers of your paper to get to
because i know it's
i know i blew it out of proportion
because when i first read it
it was a big deal but maybe you can help
me bring it back into some sort of
scale as to what what was your deep hope
here yeah it's interesting
no i mean i didn't i i don't i also
don't mean to suggest that you got it
wrong
or anything with i think the proportion
you're finding
interpretations that are consistent with
what i said you know it's not like it's
not there
but um and it's funny that if i'm trying
to
if it seems like i'm trying to put ad
you know
to put things in proportion or ground it
in any way it's kind of funny because i

feel like other
others are trying to are trying to do
that like
you know relate this to embodiment and
activism and things much more concrete
and i
i keep trying to keep it abstract but um
but anyway the main
i don't know what the hope was here i
think
um
yeah i'm not sure i i think i think i
just really wanted to point to the
possib
it was really about this identity thesis
right which i'm not even sure that
priyary i agree with by the way like i'm
not sure that identity theory is the way
to go in understanding the
mind's relation to the brain i'm kind of
like not really a reductionist
but um i wanted to i it seemed like
there was a possibility of of uh
articulating a like
quantitative version of that right
using the tools that we've developed
over the past several
you know a couple decades that lewis
didn't have access to
um and i just really i i just love david
lewis as a philosopher i think he had
some amazing ideas and i
i wanted to uh and it just seemed to
it's not so much about that i wanted to
to show anything in particular it just
seemed to me that there was a
natural convergence here that that could
be spelled out

so um that's kind of what i was trying
to do
thank you so steven with a raised hand
and then anyone else
who raises their hand and also in the
last half hour or so
anyone who wants to ask a question in
the live chat so stephen then adam
um i was just sort of tying in a bit
with what
was also said by adam uh around
the the hippocampus so
you've got the place cells in the
hippocampus which gives the
there's some way to bring the
environment in potentially
in this semantic i like i've not heard
it thought about as semantics but a
semantics of place
if that makes sense so now we've got our
environment
external states potentially being
brought in
in an attractive way you've got
action states could be the grid cells
with the whatness of what's out there
and the sensory states is this lower
level flux that's coming into the bottom
of the brain
and then the the generative model
would be the the the higher level
kind of well i'll be interested what the
generative model is
in all of that that if that makes sense
if if we were to play in that
in that sandpit it would it would give
that way to bridge
that i'm a big fan of the hippocampus

but
let's just uh but i i'd love to see how
it all fits in with some of these pieces
but
i'm just curious around if that if that
kind of
uh role for these grid and place cells
could be quite
useful to to do this tractability
between higher and
lower level states here alex
if you have any thoughts there otherwise
i'll go to a question from the chat and
then
adam um yeah i mean i guess just really
briefly it seems to me that if we if we
are using these
these place cells in in this sort of
we've leveraged this thing that was
originally meant for
physical spatial navigation and we've
transposed it to like
conceptual spatial navigation
essentially that's
really cool i could see it being the
sort of the thing that conditions
expectations for the sensory motor stuff
um and sort of it would just sort of
define the generative model i guess the
dynamics at that highest
level would be the thing that sort of
sets the
the set points for things but i think
adam might have more to say
so just to the question which is also
for adam says
uh thanks adam did mention that the
transition process

probably involves quantum tunneling may
i ask if he thinks
the same for the transition from mind
meandering
back to focused attention so adam feel
free to take up that question as well as
anything else you'd like to add and then
anyone can raise their hand
oh boy um i mean i guess like
um there's a sense in which like one of
the quantum tunneling events
i guess i would have in mind so like
there'll be like these uh
regimes of sense making where the
hippocampal system will lay down
a given tiling of the hippocampal system
and collaboration the anterinal cortex
of the grid cells will
lay down this tiling of some domain
within which
you're doing this modeling with where
that you can get basically
the relationship among these bump
attractors you can get this like
structure the syntax and through the
pointers the kind of synaptics
uh and also with its themselves having a
semantics of space maybe
um just in the relation but the
so i guess the quantum tunnel like so so
but then
if you get enough prediction error from
the overall system this seems to trigger
these resetting events
where you'll break up the frame and then
you'll do a new tiling and it's kind of
like a re-grip
on the active inference it's like a new

kind of engagement
a a new set of policies that could fit
within this different
uh conceptualization of the safe space
you're working with
and so um these events of remapping i
guess you can maybe think of them as
like the quantum tunneling
it's like you're just you're not doing
um
i don't know if this works but it's like
they're called like um
some people come like levy flights like
an animal is like foraging in a patch
and then it's like and i'll just like
jump to a greater patch once it's like
not getting enough
and so it'd be like you're kind of like
imagining in this place
and you're simulating this place and
then you're like okay a little bit bored
not quite the thing and then now you're
imagining something else then you're
imagining something else i guess you
could think of those as like
quantum tunneling like you're not just
like working within one
regime of simulation just fixing that
thing you're moving from here to here to
here
but uh the question i had when i raised
my hand was um
so there's a sense in which you might
think of like
predictive coding style models of cortex
i know it's like more complicated than
predictive coding but there's a sense in
which

something like that is true probably
true because
if you do a primarily suppressive regime
you end up inducing sparsity and if the
events that
are in the world are clocking slower
than your internal updates
you come out ahead in terms of like i
mean that's why they did it for
video coding it's just more efficient
and so i'm wondering is is that enough
to give like a psychophysical identity
mapping um
in terms of like there's a sense in
which
um you will minimize predictive coding
mechanisms so they say we can
go with just those um they should
minimize
uh they should minimize activity
yeah um yeah that's that's actually kind
of where i'm coming from
with this paper um in a
to a large extent i was thinking about
yeah
predictive coding as a way of
understanding neural networks
um so i think although this perspective
can
uh can encompass you know active
inference and such i think
that's sort of the core of it is that
the
brain is literally trying to yeah to um
right sparsity as you said right induced
varsity minimize
activity only pass forward signals that
need to be passed forward and that means

that you're sort of quelling activity
that doesn't get passed forward so i
think i don't know if that's strong
enough to get you an identity
but i think it's uh it's consistent with
the identity idea and it's um
i think it's a large part of the
inspiration for me
awesome stephen with a raised hand and
then anyone else
yeah if you could just speak to how you
see that predictive coding predictive
processing
be more or less useful at certain times
um
as the sort of the the the discourse
and when maybe active inference is not
as needed at certain times you know
maybe
thinking in applied context but um i'll
just be curious
yeah um it seems
it seems to me that if you like that if
you want a
like the the the main advantage of
active infants practically speaking to
me seems to be
like modeling uh
like explicitly how how policies
actions are chosen like decision making
right based on
um how you expect
your actions to change the way states
evolve
so i think that that happens implicitly
in a well-trained predictive coding
architecture
right i and i i see i don't see any

fundamental theoretical advantage to
active inference there in terms of
explaining how things work
on like a in principle level but
that said it's really hard to like just
train a recurrent neural network or
whatever to
to do what you want to do and then
interpret its states and so on so if you
have an idea of a generative model
like of what the agent's generative
model should look like
then it makes much more sense to write
it down a priori and
you know try to build an active
inference model
than to just learn it from data um so
that's as close to maybe a
applied setting as i could get but i
hope that sort of speaks to it
yeah let me let me try to get another
view on that from you
alex where would you put active
inference
in relationship to supervised and
unsupervised learning
for those who might be less familiar
with those topics
when you're talking about learning from
the data versus a generative model
where does active inference fit into
that yeah that's a great question
um it's it's a bit
it's a bit tricky because there's this
issue of preference learning that's kind
of not settled
in the you know like as far as uh how
how much of the future of of real

uh biological agents uh learning do we
think that is
but like um so to me the most natural
way of thinking of
of unsupervised learning happening in
active infants is that you've got sort
of a
um you've got a phenotype right some
some aspects of which are going to be
immutable over
relevant time scales but some aspects of
which will change
right so we're talking about synaptic
weights you could think of that as
an extended part of your phenotype and
that will change over
you know time scales of perceptual
learning and and during the life
organism's lifetime and such so like um
that that is i think is neatly
you can categorize that as a form of
unsupervised learning um
but that's also not really
uh again it's not it's not there are
people who
are active in sort of practitioners who
don't really
uh think that preference learning is a
thing that happens or
um yeah it's not clear
what role it plays um but but but
overall i mean
uh well there's also there's also gotta
be a role for um
something like uh supervised learning
seems to me the closest thing
that we have to that in real systems is
is like

uh reinforcement in a way so like
uh and that's certainly gotta play a
role that's complementary to
unsupervised learning but i think
um
i don't know this this answer is kind of
i'm sorry uh
it's uh i'm happy to to
to take it up more i'm just not sure
what like i don't know how to
how to how to tease apart the you know
the the parts of active inference that
should be understood in terms of
unsupervised learning versus just like
learning in general and how much of it i
i suspect a lot of learning in general
has
got to be unsupervised and i think
active inference
is sort of uh
naturally interpreted it's predictive
coding certainly is is
is an unsupervised learning paradigm and
to the extent that active influence
reduces to that
it is too great thank you
and these are definitely all topics
we're always trying to
think clearer about and communicate
clearer about so stephen
and then adam
i think this is a useful conversation i
know it's a bit off it's pretty on the
spot a little bit but i think as a group
it's quite useful
this question of um when a generative
model comes into play
so with reinforcement learning i suppose

there's a generative model
in relation to like an induction an
inductive reasoning in terms of the goal
like you're trying to narrow the gap
between the target that you're
reinforcing the reward
and reducing the gap whereas there's
this abductive potential
in active inference to
and i wonder whether in the in the
population
or the population of the brain's
attention
i'm sort of wondering if place cells and
the sort of more granular parts of the
lower levels of the neocortex or i'm not
sure if that's the right
term are basically that kind of
um more
unsupervised very unsupervised very
abductive kind of potentially
ways of engaging and then these grid
cells as they get more populated with
what things are
become kind of more top down maybe more
let us go back into say a more
reinforcement learning or supervised
learning because like
and then you you don't forage as much so
to speak you uh
you just go straight to the point which
probably i should do now
and let someone else
thanks steven um alex if you want to add
something otherwise we'll go to adam
um yeah no just to say i mean that kind
of makes sense to me i think
i think you you need i think you need

unsupervised learning for
representation learning of whatever
isn't innate
right and then once you have that
framework then
then you can start to think about like
reward like meaningful
you know interpretations of sensory
stimuli that you could associate with
rewards so
that's all saved for now cool adam
um a quick uh com a question but a quick
comment on what stephen said um
i mean interestingly about hippocampal
system is there does seem to be like
this
like very contrastive thing to it um
in terms of like comparing between
current states and then
imagine states and like at different
like duty cycles of
theta you'll be seeing there seems to be
this comparison operation going on so it
could be very much like this
higher level highest level supervisor in
this like
uh matroyshka dolls of like
quasi-homuncular
things that we're thinking of um but um
the the question i i was wondering wait
there's people who don't believe
preference learning is a thing
oh i shouldn't i shouldn't have put
anyone on the spot it's not that
it's necessary they don't believe it's a
thing it's that it's not clear that we
need to invoke it to explain
behavior behavioral things in at least

in
in many cases um so i feel like i'm
gonna
i'm gonna be speaking for for people who
are colleagues of mine who i don't wanna
i don't wanna misrepresent their views
so i'll shut up about it but
there's definitely controversy about the
extent of which it happens i suppose
i mean i i'm wondering i guess like to
what ex like
so like there's things like uh like
metal learning
uh principles where like you have like
you're figuring out like new ways of
approaching policies so you're like
of realizing goals and you're building
up these like
policies like an exploratory fashion as
part of your phenotypic plasticity
and um but i guess there is a sense in
which like
if we're like for subscribing to the
free energy principle
you can only prefer one thing and that
thing never changes and everything
else is just instrumental it just you
could just only prefer
to be basically agent smith for the
entire world
you could like my pattern me me me
um right but you know ultimately
this should be like we in terms of like
the
evolutionary favorable equilibria but it
seems like
i guess there is a sense in which that
never changes

yeah i mean i just
so i say one thing to like to try to
make
defend the people that i i've been
talking about um but i say preference
is controversy about preference learning
i mean there's clearly something that
that looks like preference learning
right it's just a question of whether
you're really learning preferences or
you're just learning
new associations with with the things
that you prefer
and the complete those two processes are
conceptually distinct they might lead to
similar
results but uh yeah but the point you're
raising is interesting that is that
really your preference is like you right
literally
your phenotype so well yeah i have a
question for you
yeah that's exactly where i was going to
go could there not be a cultural
scaffolding that's developed over
ecological
and evolutionary time that values
diversity of perspective
and so yes it is true that at different
scales some type of active inference
could be happening but
i don't think that agent smith would be
the only attractor at the bottom of that
bull
it'd be more like a uh maybe a plate
with a lot of different wells a lot of
different
kinds of food that could be converged

upon
why would it have to be converging on
one specific
um especially hollywood represented
version
of what efficiency is
just a thought but yeah i don't think
agent smith would win
like villains usually lose
you know steven
well this this could be though when you
get into that non-equilibrium
equilibrium dynamic is like
once you get a dictator who establishes
an equilibrium
that's an energy equilibrium which is
that's going to be more powerful than
the more subtle
organic non-equilibrium states it just
dominates at times and if there is made
in equivalence
in some near or distant future between
actual energy usage
and governance for example some sort of
crypto
system with energy and voting and
information
and finances all being mixed together
it'll be a
enormous nexus of power and maybe we'll
be able to use these kinds of frameworks
to
describe them stephen
and also going the other way then if you
this is my belief this is
without is i i strongly believe that to
create
community engagement in meaningful ways

you have to create
the the container um
for those more subtle non-equilibrium
states to have
a chance to breathe um this this woman
ariane munchkin she's a she's a theater
director in
france she talks about don't crush the
butterfly
she will tell the actors don't crush the
butterfly the butterfly is like
the idea so don't grab hold of it
because you'll crush it
so it's quite a nice metaphor you have
to hold it lightly
and allow it to and there's there's a
certain
because equilibrium dynamics or someone
who comes in and basically starts
shouting at people
will always over dominate someone who's
trying to do
some let's all come together and find
each other's plurality
so anyway that's that sort of dynamic
sort of ties in with what you're saying
cool so i flipped to 49 where we have
our usual
set of just closing discussions so in
these last
10 minutes we'll just be remarking on
what we thought was interesting or
important from the conversation
what would we like to pursue next in
terms of our own
work or alex in terms of your research
i'm sure we're also
all curious as well about how you apply

these ideas
in terms of your you know whatever
non-proprietary
applications you think that this bears
on
if none then that's its own interesting
claim or maybe there is something that
this helps us
do in terms of practice um
what are the next steps here maybe alex
first and then anyone else can give a
thought
uh sure so i mean i could address these
questions on the slide or
yeah let's let's yeah let's hear your
question your answers to these questions
on the slide
so i mean this also speaks to what you
just asked but i think
really i think that this is just a way
of thinking about things um
that the main thing i think it enables
is just allowing you to in to infer
from what you know about cognition or
the cognitive states of some system
including yourself
to physical states right if it's true
then
then for example yeah you can say well
i'm depressed there's not a lot going on
um you know i'm i'm uh i'm ruminating a
lot
uh you can infer that there's something
physically happening
that's mirrors that process and
you can take either a physical or a
psychological
approach to solving any in your body

right in theory
now there's a lot of working out to be
done of how that would work in specific
cases but that's the kind of
like basically just sort of smoothing
the path for inference from physical to
psychological states
i think is uh is the main thing that i
think is is practical about the
practically important about this
um and i'm still curious about how to
spell that out
um for specific types of states
uh like like depression desires things
like that
and um i'm also curious about whether
any of it is true at all or whether it's
just complete nonsense
big questions adam and then anyone else
who wants to give
any thought or address one of the
questions on the slide
if there's complete nonsense i'm not
sure what would make sense i'm not even
sure that what the sense would mean um
yeah uh later i would really want to
talk to you about like
the degrees of freedom and like volition
issue
and and like different ways that could
play out and also i want to drag you
into consciousness because
you you are precise my friend and you're
helping me think better
so cool thanks adam
happy too cool um
if anyone else has a thought otherwise
we can sort of

look back oh yeah steven go for it
yeah i just will say thank you for
bringing
the sort of this foundational build-up
in
into the um thermodynamics and i'd be
interested to
yeah maybe sort of see if there's any i
i did physical chemistry um
as um so i've got some background but i
i i'm
curious to see if there's any ways to um
bring in some elements of
of sort of gibbs free energy and
uh into some of this so i'll be glad to
chat about that at some time
yeah blue
so when you guys are done talking about
degrees of freedom and
agency you have to come back to the chat
because i want to hear all about this i
have
some thoughts that have been cooking
about agency related to active inference
and
other ideas also cool and
maybe dean i'd like to ask where does
the dot two
get us what do we do at the end of the
dot two we took that step
to the dot zero with yourself and blue
and i
and then we take the dot one and dot two
steps ideally with the author
especially when it's an awesome author
like alex who's engaged with the
material
before the dot one and between the two

what's our next step
you know on that staircase what do we do
after the dot two how do we make the
best
of this well
this is a sample of one so but for me
it's uh
so how do you remain stable and dynamic
at the same time
how do you move through these gradients
whether it's ascending or descending
and also keep in in the front of your
mind
how you're going to keep your grip if
you're ascending or
maintain your balance when you're
descending and for me
that's what alex's paper essentially
pointed out there is a there's a
physical piece to that
and there's a statistical piece to that
and i don't want to become a statistic
of another
old person who falls down the stairs so
how do i pull all of that together and
have it make sense
especially if i don't want to give
somebody
a training exercise if i want them to be
able to model
how to get down that descent or up that
ascent
and i can't give them a map so what do i
what do i give them what do i replace
that with
that's what alex's paper gave to me gave
me a little bit of confidence
awesome one more staircase thought

and coming back as well to the up and
down hill
so when you're going up the staircase
you'd grab onto a certain spot on the
rail and then that helps you
ratchet up and then when you're going
down the staircase
you're unlikely to grab on tight to a
rail you kind of want to loosely have
your hand
sliding down the rails so that you can
ratchet if you needed to
and stabilize but if you ratchet while
you're going down
then it's gonna prevent you from
continuing this sort of natural
descent so it just makes me think about
different modes in understanding
especially
because as we talked about with the
mountain climber it's like going up and
down hill
in their own way if the top is your goal
they're like both going downhill
so when in our process do we want to
grip the rail lightly just like steven
said not crush the butterfly
just slide along and just go with the
flow knowing that we can
stabilize when we need to and when do we
really
want to grip and ratchet because we're
working against
one energy gradient but we're working
towards our goals when we do that
so i hope all these mixed metaphors come
together
for the listeners in an enjoyable way

and alex
we really appreciate your engagement
with actinflab here and everybody else
awesome conversations yeah thanks thanks
uh blue dean daniel steven adam uh
people who were here last week also um
i'm i'm happy to be put on the spot
i'm glad this was of service to people i
think it's been a really interesting
discussion
and um thanks for thanks for having me
thanks for for all all that you do
great well everyone's always welcome and
we'll see you all in another stream
bye
[Music]